SOLVING MATRIX EQUATIONS

LINEAR ALGEBRA

prepared by:

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

WEEKLY GRADE

70%

LONG QUIZ

30%

SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Interpretations of Matrix Multiplication

Solving Vector Equations

Solving Matrix Equations

INTERPRETATIONS OF MATRIX MULTIPLICATIONS

INTERPRETATION 1

The $i\,j$ entry of C is the **dot product** of row i in A and column j in B

$$c_{ij} = \overrightarrow{a_i} \cdot \overrightarrow{b_j}$$

$$\begin{array}{ccccc} A & & B & & C \\ \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix} & \cdot & \begin{bmatrix} 3 & 2 \\ 2 & 2 \\ 1 & 1 \end{bmatrix} & = & \begin{bmatrix} 10 & \\ & \end{bmatrix} \\ 2 \times 3 & & 3 \times 2 & & 2 \times 2 \end{array}$$

$$c_{11} = \overrightarrow{a_1} \cdot \overrightarrow{b_1}$$

$$c_{11} = 10$$

The $i\,j$ entry of C is the **dot product** of row i in A and column j in B

$$\boldsymbol{c_{ij} = \overrightarrow{a_i} \cdot \overrightarrow{b_j}}$$

$$\begin{array}{ccccc} \boldsymbol{A} & & \boldsymbol{B} & & \boldsymbol{C} \\ \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix} & \cdot & \begin{bmatrix} 3 & 2 \\ 2 & 2 \\ 1 & 1 \end{bmatrix} & = & \begin{bmatrix} 10 & \\ 7 & \end{bmatrix} \\ 2 \times 3 & & 3 \times 2 & & 2 \times 2 \end{array}$$

$$c_{11} = \overrightarrow{a_1} \cdot \overrightarrow{b_1}$$

$$\boldsymbol{c_{11} = 10}$$

$$c_{21} = \overrightarrow{a_2} \cdot \overrightarrow{b_1}$$

$$\boldsymbol{c_{21} = 7}$$

The $i\,j$ entry of C is the **dot product** of row i in A and column j in B

$$\boldsymbol{c_{ij} = \overrightarrow{a_i} \cdot \overrightarrow{b_j}}$$

$$\begin{array}{ccccc} \boldsymbol{A} & & \boldsymbol{B} & & \boldsymbol{C} \\ \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix} & \cdot & \begin{bmatrix} 3 & 2 \\ 2 & 2 \\ 1 & 1 \end{bmatrix} & = & \begin{bmatrix} 10 & 9 \\ 7 & \end{bmatrix} \\ 2 \times 3 & & 3 \times 2 & & 2 \times 2 \end{array}$$

$$c_{11} = \overrightarrow{a_1} \cdot \overrightarrow{b_1}$$

$$\boldsymbol{c_{11} = 10}$$

$$c_{12} = \overrightarrow{a_1} \cdot \overrightarrow{b_2}$$

$$\boldsymbol{c_{12} = 9}$$

$$c_{21} = \overrightarrow{a_2} \cdot \overrightarrow{b_1}$$

$$\boldsymbol{c_{21} = 7}$$

The $i\,j$ entry of C is the **dot product** of row i in A and column j in B

$$\boldsymbol{c_{ij} = \overrightarrow{a_i} \cdot \overrightarrow{b_j}}$$

$$\begin{array}{ccccc} \boldsymbol{A} & & \boldsymbol{B} & & \boldsymbol{C} \\ \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix} & \cdot & \begin{bmatrix} 3 & 2 \\ 2 & 2 \\ 1 & 1 \end{bmatrix} & = & \begin{bmatrix} 10 & 9 \\ 7 & 6 \end{bmatrix} \\ 2 \times 3 & & 3 \times 2 & & 2 \times 2 \end{array}$$

$$c_{11} = \overrightarrow{a_1} \cdot \overrightarrow{b_1}$$

$$\boldsymbol{c_{11} = 10}$$

$$c_{12} = \overrightarrow{a_1} \cdot \overrightarrow{b_2}$$

$$\boldsymbol{c_{12} = 9}$$

$$c_{21} = \overrightarrow{a_2} \cdot \overrightarrow{b_1}$$

$$\boldsymbol{c_{21} = 7}$$

$$c_{22} = \overrightarrow{a_2} \cdot \overrightarrow{b_2}$$

$$\boldsymbol{c_{22} = 6}$$

INTERPRETATION 2

$$\begin{array}{ccccc} A & & B & & C \\ \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix} & \cdot & \begin{bmatrix} 3 & 2 \\ 2 & 2 \\ 1 & 1 \end{bmatrix} & = & \begin{bmatrix} & \\ & \end{bmatrix} \\ 2 \times 3 & & 3 \times 2 & & 2 \times 2 \end{array}$$

The jth column of C is the product of **matrix** A and the jth **column** of B

$$\overrightarrow{c_j} = A \cdot \overrightarrow{b_j}$$

INTERPRETATION 2

The jth column of C is the product of **matrix** A and the jth **column** of B

$$\overrightarrow{c_j} = A \cdot \overrightarrow{b_j}$$

$$\begin{array}{ccccc} A & & \overrightarrow{b_1} & & \overrightarrow{c_1} \\ \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix} & \cdot & \begin{bmatrix} 3 & \\ 2 & \\ 1 & \end{bmatrix} & = & \begin{bmatrix} 10 & \\ 7 & \end{bmatrix} \\ 2 \times 3 & & 3 \times 2 & & 2 \times 2 \end{array}$$

$$A \cdot \overrightarrow{b_1} = \overrightarrow{c_1}$$

$$\begin{bmatrix} 10 \\ 7 \end{bmatrix}$$

The jth column of C is the product of **matrix** A and the jth **column** of B

$$\overrightarrow{c_j} = A \cdot \overrightarrow{b_j}$$

$$\underset{2\times 3}{\overset{A}{\begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix}}} \cdot \underset{3\times 2}{\overset{\overrightarrow{b_2}}{\begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}}} = \underset{2\times 2}{\overset{\overrightarrow{c_2}}{\begin{bmatrix} 9 \\ 6 \end{bmatrix}}}$$

$$A \cdot \overrightarrow{b_1} = \overrightarrow{c_1} \qquad A \cdot \overrightarrow{b_2} = \overrightarrow{c_2}$$

$$\begin{bmatrix} 10 \\ 7 \end{bmatrix} \qquad \begin{bmatrix} 9 \\ 6 \end{bmatrix}$$

$$\underset{2\times 3}{\overset{A}{\begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix}}} \cdot \underset{3\times 2}{\overset{B}{\begin{bmatrix} 3 & 2 \\ 2 & 2 \\ 1 & 1 \end{bmatrix}}} = \underset{2\times 2}{\overset{C}{\begin{bmatrix} 10 & 9 \\ 7 & 6 \end{bmatrix}}}$$

The jth column of C is the product of **matrix** A and the jth **column** of B

$$\overrightarrow{c_j} = A \cdot \overrightarrow{b_j}$$

$$A \cdot \overrightarrow{b_1} = \overrightarrow{c_1} \qquad A \cdot \overrightarrow{b_2} = \overrightarrow{c_2}$$

$$\begin{bmatrix} 10 \\ 7 \end{bmatrix} \qquad \begin{bmatrix} 9 \\ 6 \end{bmatrix}$$

Given that

$$\begin{bmatrix} 4 & 5 \\ 6 & 7 \\ 8 & 9 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 13 \\ 19 \\ 25 \end{bmatrix} \qquad \begin{bmatrix} 4 & 5 \\ 6 & 7 \\ 8 & 9 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 9 \\ 13 \\ 17 \end{bmatrix} \qquad \begin{bmatrix} 4 & 5 \\ 6 & 7 \\ 8 & 9 \end{bmatrix} \begin{bmatrix} 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 15 \\ 21 \\ 27 \end{bmatrix}$$

Find

$$\begin{bmatrix} 4 & 5 \\ 6 & 7 \\ 8 & 9 \end{bmatrix} \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Given that

$$\overset{A}{\begin{bmatrix} 4 & 5 \\ 6 & 7 \\ 8 & 9 \end{bmatrix}} \overset{\overrightarrow{b_1}}{\begin{bmatrix} 2 \\ 1 \end{bmatrix}} = \overset{\overrightarrow{c_1}}{\begin{bmatrix} 13 \\ 19 \\ 25 \end{bmatrix}} \qquad \overset{A}{\begin{bmatrix} 4 & 5 \\ 6 & 7 \\ 8 & 9 \end{bmatrix}} \overset{\overrightarrow{b_2}}{\begin{bmatrix} 1 \\ 1 \end{bmatrix}} = \overset{\overrightarrow{c_2}}{\begin{bmatrix} 9 \\ 13 \\ 17 \end{bmatrix}} \qquad \overset{A}{\begin{bmatrix} 4 & 5 \\ 6 & 7 \\ 8 & 9 \end{bmatrix}} \overset{\overrightarrow{b_3}}{\begin{bmatrix} 0 \\ 3 \end{bmatrix}} = \overset{\overrightarrow{c_3}}{\begin{bmatrix} 15 \\ 21 \\ 27 \end{bmatrix}}$$

Find

$$\overset{A}{\begin{bmatrix} 4 & 5 \\ 6 & 7 \\ 8 & 9 \end{bmatrix}} \overset{B}{\begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}} = \underbrace{\overset{C}{\begin{bmatrix} 13 & 9 & 15 \\ 19 & 13 & 21 \\ 25 & 17 & 27 \end{bmatrix}}}_{ans}$$

SOLVING VECTOR EQUATIONS

Solving vector equation of the form

$$\boldsymbol{A\vec{x} = \vec{b}}$$

where,

A = matrix

$\vec{x}$ = unknown vector

$\vec{b}$ = known vector

Solve the vector equation $A\vec{x} = \vec{b}$ for **vector $\vec{x}$**, where

$$A = \begin{bmatrix} 1 & 2 & -2 \\ 3 & 4 & -5 \\ 0 & 1 & 1 \end{bmatrix} \quad \text{and} \quad \vec{b} = \begin{bmatrix} 1 \\ 7 \\ -2 \end{bmatrix}$$

$$\underset{3 \times 3}{\overset{A}{\begin{bmatrix} 1 & 2 & -2 \\ 3 & 4 & -5 \\ 0 & 1 & 1 \end{bmatrix}}} \underset{3 \times 1}{\overset{\vec{x}}{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}} = \underset{3 \times 1}{\overset{\vec{b}}{\begin{bmatrix} 1 \\ 7 \\ -2 \end{bmatrix}}}$$

Interpretation 1,

$$1x_1 + 2x_2 - 2x_3 = 1$$

Solve the vector equation $A\vec{x} = \vec{b}$ for **vector $\vec{x}$**, where

$$A = \begin{bmatrix} 1 & 2 & -2 \\ 3 & 4 & -5 \\ 0 & 1 & 1 \end{bmatrix} \quad \text{and} \quad \vec{b} = \begin{bmatrix} 1 \\ 7 \\ -2 \end{bmatrix}$$

$$\underset{3 \times 3}{\overset{A}{\begin{bmatrix} 1 & 2 & -2 \\ 3 & 4 & -5 \\ 0 & 1 & 1 \end{bmatrix}}} \underset{3 \times 1}{\overset{\vec{x}}{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}} = \underset{3 \times 1}{\overset{\vec{b}}{\begin{bmatrix} 1 \\ 7 \\ -2 \end{bmatrix}}}$$

Interpretation 1,

$$1x_1 + 2x_2 - 2x_3 = 1$$

$$3x_1 + 4x_2 - 5x_2 = 7$$

Solve the vector equation $A\vec{x} = \vec{b}$ for **vector $\vec{x}$**, where

$$A = \begin{bmatrix} 1 & 2 & -2 \\ 3 & 4 & -5 \\ 0 & 1 & 1 \end{bmatrix} \quad \text{and} \quad \vec{b} = \begin{bmatrix} 1 \\ 7 \\ -2 \end{bmatrix}$$

$$\underset{3 \times 3}{\overset{A}{\begin{bmatrix} 1 & 2 & -2 \\ 3 & 4 & -5 \\ 0 & 1 & 1 \end{bmatrix}}} \underset{3 \times 1}{\overset{\vec{x}}{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}} = \underset{3 \times 1}{\overset{\vec{b}}{\begin{bmatrix} 1 \\ 7 \\ -2 \end{bmatrix}}}$$

Interpretation 1,

$$1x_1 + 2x_2 - 2x_3 = 1$$

$$3x_1 + 4x_2 - 5x_2 = 7$$

$$0x_1 + 1x_2 + 1x_3 = -2$$

Solve the vector equation $A\vec{x} = \vec{b}$ for **vector $\vec{x}$**, where

$$A = \begin{bmatrix} 1 & 2 & -2 \\ 3 & 4 & -5 \\ 0 & 1 & 1 \end{bmatrix} \quad \text{and} \quad \vec{b} = \begin{bmatrix} 1 \\ 7 \\ -2 \end{bmatrix}$$

$$\underset{3 \times 3}{\overset{A}{\begin{bmatrix} 1 & 2 & -2 \\ 3 & 4 & -5 \\ 0 & 1 & 1 \end{bmatrix}}} \underset{3 \times 1}{\overset{\vec{x}}{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}} = \underset{3 \times 1}{\overset{\vec{b}}{\begin{bmatrix} 1 \\ 7 \\ -2 \end{bmatrix}}}$$

System of Linear Equations,

$$1x_1 + 2x_2 - 2x_3 = 1$$

$$3x_1 + 4x_2 - 5x_2 = 7$$

$$0x_1 + 1x_2 + 1x_3 = -2$$

Augmented Matrix,

$$\begin{array}{c} \begin{array}{cccc} x_1 & x_2 & x_3 & \end{array} \\ \begin{bmatrix} 1 & 2 & -2 & 1 \\ 3 & 4 & -5 & 7 \\ 0 & 1 & 1 & -2 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \end{array}$$

Augmented Matrix,

$$\begin{array}{cc} \begin{array}{cccc} x_1 & x_2 & x_3 & \end{array} & \\ \begin{bmatrix} 1 & 2 & -2 & 1 \\ 3 & 4 & -5 & 7 \\ 0 & 1 & 1 & -2 \end{bmatrix} & \begin{array}{c} r_1 \\ r_2 \\ r_3 \end{array} \end{array}$$

Augmented Matrix,

$$\begin{array}{cc} \begin{array}{cccc} x_1 & x_2 & x_3 & \end{array} & \\ \begin{bmatrix} 1 & 2 & -2 & 1 \\ 3 & 4 & -5 & 7 \\ 0 & 1 & 1 & -2 \end{bmatrix} & \begin{array}{l} r_1 \\ r_2 \leftarrow -3r_1 + r_2 \\ r_3 \end{array} \end{array}$$

$$\begin{array}{cc} \begin{array}{cccc} x_1 & x_2 & x_3 & \end{array} & \\ \begin{bmatrix} 1 & 2 & -2 & 1 \\ 0 & -2 & -1 & 4 \\ 0 & 1 & 1 & -2 \end{bmatrix} & \begin{array}{l} r_1 \\ r_2 \\ r_3 \leftarrow 2r_3 + r_2 \end{array} \end{array}$$

$$\begin{array}{cc} \begin{array}{cccc} x_1 & x_2 & x_3 & \end{array} & \\ \begin{bmatrix} 1 & 2 & -2 & 1 \\ 0 & -2 & -1 & 4 \\ 0 & 0 & 1 & 0 \end{bmatrix} & \begin{array}{l} r_1 \leftarrow r_1 + r_2 \\ r_2 \\ r_3 \end{array} \end{array}$$

$$\begin{array}{cc} \begin{array}{cccc} x_1 & x_2 & x_3 & \end{array} & \\ \begin{bmatrix} 1 & 0 & -3 & 5 \\ 0 & -2 & -1 & 4 \\ 0 & 0 & 1 & 0 \end{bmatrix} & \begin{array}{l} r_1 \\ r_2 \leftarrow r_2 + r_3 \\ r_3 \end{array} \end{array}$$

Augmented Matrix,

$$\begin{array}{cccc} x_1 & x_2 & x_3 & \\ \end{array}$$

$$\begin{bmatrix} 1 & 0 & -3 & 5 \\ 0 & -2 & -1 & 4 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{array}{l} r_1 \\ r_2 \leftarrow r_2 + r_3 \\ r_3 \end{array}$$

Augmented Matrix,

$$\begin{array}{cc} & \begin{array}{cccc} x_1 & x_2 & x_3 & \end{array} \\ & \begin{bmatrix} 1 & 0 & -3 & 5 \\ 0 & -2 & 0 & 4 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{array} \begin{array}{l} r_1 \\ r_2 \\ r_3 \end{array}$$

$$\begin{array}{cc} & \begin{array}{cccc} x_1 & x_2 & x_3 & \end{array} \\ & \begin{bmatrix} 1 & 0 & -3 & 5 \\ 0 & -2 & -1 & 4 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{array} \begin{array}{l} r_1 \\ r_2 \leftarrow r_2 + r_3 \\ r_3 \end{array}$$

Augmented Matrix,

$$\begin{array}{cccc} x_1 & x_2 & x_3 & \\ \end{array}$$
$$\begin{bmatrix} 1 & 0 & -3 & 5 \\ 0 & -2 & 0 & 4 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{array}{l} r_1 \leftarrow 3r_3 + r_1 \\ r_2 \leftarrow r_2/-2 \\ r_3 \end{array}$$

RREF,

$$\begin{array}{cccc} x_1 & x_2 & x_3 & \\ \end{array}$$
$$\begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{array}{l} r_1 \\ r_2 \\ r_3 \end{array}$$

Solve the vector equation $A\vec{x} = \vec{b}$ for vector $\vec{x}$,

where

$$A = \begin{bmatrix} 1 & 2 & -2 \\ 3 & 4 & -5 \\ 0 & 1 & 1 \end{bmatrix} \quad \text{and} \quad \vec{b} = \begin{bmatrix} 1 \\ 7 \\ -2 \end{bmatrix}$$

$$\underset{3 \times 3}{\overset{A}{\begin{bmatrix} 1 & 2 & -2 \\ 3 & 4 & -5 \\ 0 & 1 & 1 \end{bmatrix}}} \underset{3 \times 1}{\overset{\vec{x}}{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}} = \underset{3 \times 1}{\overset{\vec{b}}{\begin{bmatrix} 1 \\ 7 \\ -2 \end{bmatrix}}}$$

RREF,

$$\begin{array}{c} \begin{array}{cccc} x_1 & x_2 & x_3 & \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \end{array}$$

$$\vec{x} = \underline{\begin{bmatrix} 5 \\ -2 \\ 0 \end{bmatrix}}$$

ans

SOLVING MATRIX EQUATIONS

Solving matrix equation of the form

$$\boldsymbol{AX} = \boldsymbol{B}$$

where,

A = known matrix

X = unknown matrix

B = known matrix

Solve the matrix equation $\boldsymbol{AX} = \boldsymbol{B}$ for matrix X ,

where

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 7 & -10 & 5 \\ 1 & -2 & 7 \end{bmatrix}$$

$$\underset{2 \times 2}{\overset{A}{\begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}}} \underset{2 \times 3}{\overset{X}{\begin{bmatrix} \quad & \quad & \quad \\ \quad & \quad & \quad \end{bmatrix}}} = \underset{2 \times 3}{\overset{B}{\begin{bmatrix} 7 & -10 & 5 \\ 1 & -2 & 7 \end{bmatrix}}}$$

Solve the matrix equation $\boldsymbol{AX} = \boldsymbol{B}$ for matrix X ,

where

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 7 & -10 & 5 \\ 1 & -2 & 7 \end{bmatrix}$$

$$\begin{array}{ccc} A & X & B \\ \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} & \begin{bmatrix} \overrightarrow{x_1} & \overrightarrow{x_2} & \overrightarrow{x_3} \end{bmatrix} & = \begin{bmatrix} 7 & -10 & 5 \\ 1 & -2 & 7 \end{bmatrix} \\ 2 \times 2 & 2 \times 3 & 2 \times 3 \end{array}$$

Interpretation 2,

$$A \cdot \overrightarrow{x_1} = \overrightarrow{b_1}$$

$$\begin{array}{cl} \begin{bmatrix} 1 & 2 & 7 \\ 3 & 2 & 1 \end{bmatrix} & \begin{array}{l} r_1 \\ r_2 \leftarrow -3r_1 + r_2 \end{array} \end{array}$$

$$\begin{array}{cl} \begin{bmatrix} 1 & 2 & 7 \\ 0 & -4 & -20 \end{bmatrix} & \begin{array}{l} r_1 \\ r_2 \leftarrow r_2/-4 \end{array} \end{array}$$

$$\begin{array}{cl} \begin{bmatrix} 1 & 2 & 7 \\ 0 & 1 & 5 \end{bmatrix} & \begin{array}{l} r_1 \leftarrow -2r_2 + r_1 \\ r_2 \end{array} \end{array}$$

$$\begin{array}{cl} \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 5 \end{bmatrix} & \begin{array}{l} r_1 \\ r_2 \end{array} \end{array}$$

Solve the matrix equation $\boldsymbol{AX} = \boldsymbol{B}$ for matrix X ,

where

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 7 & -10 & 5 \\ 1 & -2 & 7 \end{bmatrix}$$

$$\begin{array}{ccc} A & X & B \\ \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} & \begin{bmatrix} -3 & \overrightarrow{x_2} & \overrightarrow{x_3} \\ 5 & & \end{bmatrix} & = \begin{bmatrix} 7 & -10 & 5 \\ 1 & -2 & 7 \end{bmatrix} \\ 2 \times 2 & 2 \times 3 & 2 \times 3 \end{array}$$

Interpretation 2,

$$A \cdot \overrightarrow{x_1} = \overrightarrow{b_1}$$

$$\begin{bmatrix} 1 & 2 & 7 \\ 3 & 2 & 1 \end{bmatrix} \begin{array}{l} r_1 \\ r_2 \leftarrow -3r_1 + r_2 \end{array}$$

$$\begin{bmatrix} 1 & 2 & 7 \\ 0 & -4 & -20 \end{bmatrix} \begin{array}{l} r_1 \\ r_2 \leftarrow r_2/-4 \end{array}$$

$$\begin{bmatrix} 1 & 2 & 7 \\ 0 & 1 & 5 \end{bmatrix} \begin{array}{l} r_1 \leftarrow -2r_2 + r_1 \\ r_2 \end{array}$$

$$\begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 5 \end{bmatrix} \begin{array}{l} r_1 \\ r_2 \end{array}$$

Solve the matrix equation $\boldsymbol{AX} = \boldsymbol{B}$ for matrix X ,

where

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 7 & -10 & 5 \\ 1 & -2 & 7 \end{bmatrix}$$

$$\begin{array}{ccc} A & X & B \\ \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} & \begin{bmatrix} -3 & \overrightarrow{x_2} & \overrightarrow{x_3} \\ 5 & & \end{bmatrix} & = \begin{bmatrix} 7 & -10 & 5 \\ 1 & -2 & 7 \end{bmatrix} \\ 2 \times 2 & 2 \times 3 & 2 \times 3 \end{array}$$

Interpretation 2,

$$A \cdot \overrightarrow{x_2} = \overrightarrow{b_2}$$

$$\begin{bmatrix} 1 & 2 & -10 \\ 3 & 2 & -2 \end{bmatrix} \begin{array}{l} r_1 \\ r_2 \leftarrow -3r_1 + r_2 \end{array}$$

$$\begin{bmatrix} 1 & 2 & -10 \\ 0 & -4 & 28 \end{bmatrix} \begin{array}{l} r_1 \\ r_2 \leftarrow r_2/-4 \end{array}$$

$$\begin{bmatrix} 1 & 2 & -10 \\ 0 & 1 & -7 \end{bmatrix} \begin{array}{l} r_1 \leftarrow -2r_2 + r_1 \\ r_2 \end{array}$$

$$\begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & -7 \end{bmatrix} \begin{array}{l} r_1 \\ r_2 \end{array}$$

Solve the matrix equation $\boldsymbol{AX} = \boldsymbol{B}$ for matrix X ,

where

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 7 & -10 & 5 \\ 1 & -2 & 7 \end{bmatrix}$$

$$\begin{array}{ccc} A & X & B \\ \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} & \begin{bmatrix} -3 & 4 & \overrightarrow{x_3} \\ 5 & -7 & \end{bmatrix} & = \begin{bmatrix} 7 & -10 & 5 \\ 1 & -2 & 7 \end{bmatrix} \\ 2 \times 2 & 2 \times 3 & 2 \times 3 \end{array}$$

Interpretation 2,

$$A \cdot \overrightarrow{x_2} = \overrightarrow{b_2}$$

$$\begin{array}{cl} \begin{bmatrix} 1 & 2 & -10 \\ 3 & 2 & -2 \end{bmatrix} & \begin{array}{l} r_1 \\ r_2 \leftarrow -3r_1 + r_2 \end{array} \end{array}$$

$$\begin{array}{cl} \begin{bmatrix} 1 & 2 & -10 \\ 0 & -4 & 28 \end{bmatrix} & \begin{array}{l} r_1 \\ r_2 \leftarrow r_2/-4 \end{array} \end{array}$$

$$\begin{array}{cl} \begin{bmatrix} 1 & 2 & -10 \\ 0 & 1 & -7 \end{bmatrix} & \begin{array}{l} r_1 \leftarrow -2r_2 + r_1 \\ r_2 \end{array} \end{array}$$

$$\begin{array}{cl} \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & -7 \end{bmatrix} & \begin{array}{l} r_1 \\ r_2 \end{array} \end{array}$$

Solve the matrix equation $\boldsymbol{AX} = \boldsymbol{B}$ for matrix X ,

where

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 7 & -10 & 5 \\ 1 & -2 & 7 \end{bmatrix}$$

$$\begin{array}{ccc} A & X & B \\ \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} & \begin{bmatrix} -3 & 4 & \overrightarrow{x_3} \\ 5 & -7 & \end{bmatrix} & = \begin{bmatrix} 7 & -10 & 5 \\ 1 & -2 & 7 \end{bmatrix} \\ 2 \times 2 & 2 \times 3 & 2 \times 3 \end{array}$$

Interpretation 2,

$$A \cdot \overrightarrow{x_3} = \overrightarrow{b_3}$$

$$\begin{bmatrix} 1 & 2 & 5 \\ 3 & 2 & 7 \end{bmatrix} \begin{matrix} r_1 \\ r_2 \leftarrow -3r_1 + r_2 \end{matrix}$$

$$\begin{bmatrix} 1 & 2 & 5 \\ 0 & -4 & -8 \end{bmatrix} \begin{matrix} r_1 \\ r_2 \leftarrow r_2/-4 \end{matrix}$$

$$\begin{bmatrix} 1 & 2 & 5 \\ 0 & 1 & 2 \end{bmatrix} \begin{matrix} r_1 \leftarrow -2r_2 + r_1 \\ r_2 \end{matrix}$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{matrix} r_1 \\ r_2 \end{matrix}$$

Solve the matrix equation $\boldsymbol{AX} = \boldsymbol{B}$ for matrix X ,

where

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 7 & -10 & 5 \\ 1 & -2 & 7 \end{bmatrix}$$

$$\begin{array}{ccc} A & X & B \\ \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} & \begin{bmatrix} -3 & 4 & 1 \\ 5 & -7 & 2 \end{bmatrix} & = \begin{bmatrix} 7 & -10 & 5 \\ 1 & -2 & 7 \end{bmatrix} \\ 2 \times 2 & 2 \times 3 & 2 \times 3 \end{array}$$

Interpretation 2,

$$A \cdot \overrightarrow{x_3} = \overrightarrow{b_3}$$

$$\begin{bmatrix} 1 & 2 & 5 \\ 3 & 2 & 7 \end{bmatrix} \begin{array}{l} r_1 \\ r_2 \leftarrow -3r_1 + r_2 \end{array}$$

$$\begin{bmatrix} 1 & 2 & 5 \\ 0 & -4 & -8 \end{bmatrix} \begin{array}{l} r_1 \\ r_2 \leftarrow r_2/-4 \end{array}$$

$$\begin{bmatrix} 1 & 2 & 5 \\ 0 & 1 & 2 \end{bmatrix} \begin{array}{l} r_1 \leftarrow -2r_2 + r_1 \\ r_2 \end{array}$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{array}{l} r_1 \\ r_2 \end{array}$$

Solve the matrix equation $\boldsymbol{AX} = \boldsymbol{B}$ for **matrix X** ,

where

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 7 & -10 & 5 \\ 1 & -2 & 7 \end{bmatrix}$$

$$\begin{array}{ccc} A & X & B \\ \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} & \begin{bmatrix} -3 & 4 & 1 \\ 5 & -7 & 2 \end{bmatrix} & = \begin{bmatrix} 7 & -10 & 5 \\ 1 & -2 & 7 \end{bmatrix} \\ 2 \times 2 & 2 \times 3 & 2 \times 3 \end{array}$$

Interpretation 2,

$$A \cdot \overrightarrow{x_3} = \overrightarrow{b_3}$$

$$\begin{bmatrix} 1 & 2 & 5 \\ 3 & 2 & 7 \end{bmatrix} \begin{array}{l} r_1 \\ r_2 \leftarrow -3r_1 + r_2 \end{array}$$

$$\begin{bmatrix} 1 & 2 & 5 \\ 0 & -4 & -8 \end{bmatrix} \begin{array}{l} r_1 \\ r_2 \leftarrow r_2/-4 \end{array}$$

$$\begin{bmatrix} 1 & 2 & 5 \\ 0 & 1 & 2 \end{bmatrix} \begin{array}{l} r_1 \leftarrow -2r_2 + r_1 \\ r_2 \end{array}$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{array}{l} r_1 \\ r_2 \end{array}$$

Solve the matrix equation $\boldsymbol{AX} = \boldsymbol{B}$ for **matrix X** ,

where

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 7 & -10 & 5 \\ 1 & -2 & 7 \end{bmatrix}$$

$$\begin{array}{ccc} A & X & B \\ \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} & \begin{bmatrix} -3 & 4 & 1 \\ 5 & -7 & 2 \end{bmatrix} & = \begin{bmatrix} 7 & -10 & 5 \\ 1 & -2 & 7 \end{bmatrix} \\ 2 \times 2 & 2 \times 3 & 2 \times 3 \end{array}$$

Interpretation 2,

$$\begin{bmatrix} 1 & 2 & 7 & -10 & 5 \\ 3 & 2 & 1 & -2 & 7 \end{bmatrix} \begin{array}{l} r_1 \\ r_2 \leftarrow -3r_1 + r_2 \end{array}$$

$$\begin{bmatrix} 1 & 2 & 7 & -10 & 5 \\ 0 & -4 & -20 & 28 & -8 \end{bmatrix} \begin{array}{l} r_1 \\ r_2 \leftarrow r_2/-4 \end{array}$$

$$\begin{bmatrix} 1 & 2 & 7 & -10 & 5 \\ 0 & 1 & 5 & -7 & 2 \end{bmatrix} \begin{array}{l} r_1 \leftarrow -2r_2 + r_1 \\ r_2 \end{array}$$

$$\begin{bmatrix} 1 & 0 & -3 & 4 & 1 \\ 0 & 1 & 5 & -7 & 2 \end{bmatrix} \begin{array}{l} r_1 \\ r_2 \end{array}$$

SEATWORK